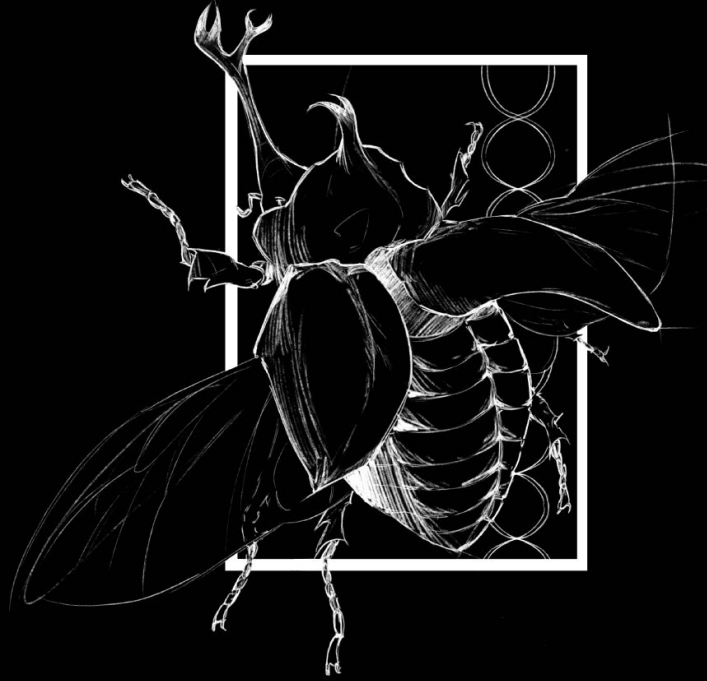




Does Habitat Type Influence Sexual Size Dimorphism in Dragonflies? A Phylogenetic Comparative Approach

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Background

The difference in body size between males and females of the same species, called sexual size dimorphism (SSD), is a widespread phenomenon across the animal kingdom. Insects especially show a high level of SSD diversity across clades. In Odonata (dragonflies and damselflies), females are frequently larger than males, a pattern thought to reflect the energetic demands of reproduction. Larger females can produce more eggs and are therefore favored by natural selection. Body size is also directly tied to reproductive success in this group, with larger males often outcompeting rivals for mates and territories.

Despite this general understanding, the ecological drivers of variation in SSD across species within Odonata are unclear. Freshwater habitats can be divided into lentic systems (standing water such as lakes and ponds) and lotic systems (flowing water such as rivers and streams). These two habitat types impose fundamentally different ecological pressures on their inhabitants. Lotic environments are characterized by stronger and more consistent currents, which may intensify competition for prime oviposition and basking sites, leading to stronger sexual selection pressures on body size. In contrast, lentic habitats have been associated with higher rates of speciation in dragonflies, potentially reflecting greater chances for reproductive isolation and divergence.

We predict that lotic species will exhibit greater female-biased SSD than lentic species, reflecting the stronger sexual selection pressures associated with that environment where competition for territories and oviposition sites is more intense.

Study Design & AI

A time-calibrated phylogeny of dragonflies was obtained from Letsch et al. (2016), which was published alongside habitat classification data for all included species. After matching species names between the tree and trait datasets, 54 dragonfly species were retained for analysis.

Male and female body length data (cm) were sourced from Córdoba-Aguilar (2008). SSD was calculated for each species as $\log(\text{female body length} / \text{male body length})$. Under this formula, a positive SSD value indicates that females are larger than males, while a negative value indicates that males are larger than females.

Lentic and lotic habitat classifications were taken from Appendix S4 of Letsch et al. (2016), which provided species-level habitat coding based on larval habitat use. Of the 54 species retained, 35 were classified as lentic and 19 as lotic.

Claude (Anthropic) was used throughout this project to assist with R scripting, data cleaning, merging of datasets, and figure production.

Model Fitting

A phylogenetic Analysis of Covariance (ANCOVA) was fitted using the `pgl()` function in the `caper` package in R. Phylogenetic ANCOVA is an extension of standard ANCOVA that incorporates the evolutionary relationships among species into the statistical model by including the phylogenetic tree. Because closely related species tend to resemble one another regardless of ecology, treating species as independent data points inflates the effective sample size and can produce false positives. By including the phylogenetic tree, the model accounts for shared ancestry and is able to concretely isolate the true effect of habitat type on SSD.

The degree of phylogenetic signal in the data was quantified using Pagel's lambda, estimated by maximum likelihood. Lambda ranges from 0 (species are effectively independent) to 1 (traits evolved under Brownian motion, meaning closely related species are very similar). The model formula was: $\text{SSD} \sim \text{habitat}$

Results

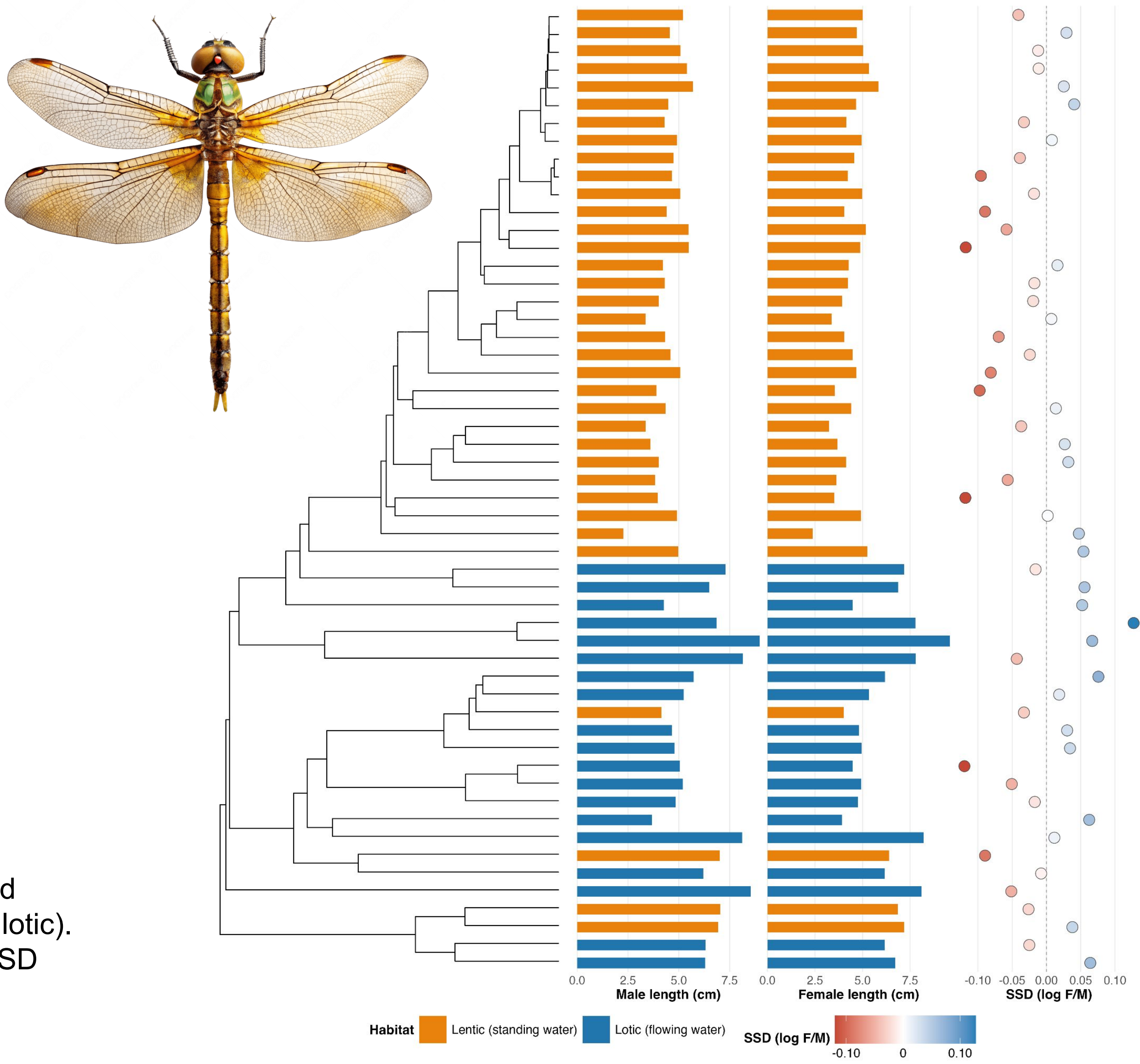
Pagel's lambda was estimated at 0.583, indicating a moderate phylogenetic signal in the data. This confirms that closely related dragonfly species tend to have more similar SSD than distantly related species, and that the phylogenetic tree inclusion was necessary and appropriate.

Habitat type trended toward significance as a predictor of SSD ($p = 0.084$), with lotic species showing greater female-biased dimorphism than lentic species. While this result does not reach the conventional threshold of $p = 0.05$, the consistent direction of the effect across analyses suggests a real biological pattern that may be detectable with a larger sample.

Both sexes were substantially larger in lotic environments than in lentic environments. Mean male body length was 6.14 cm in lotic species compared to 4.67 cm in lentic species. Mean female body length was 6.23 cm in lotic species compared to 4.56 cm in lentic species. Females showed a slightly larger absolute increase in body size between habitats (1.67 cm) than males (1.47 cm), suggesting that females grow proportionally larger than males in lotic environments, which drives the observed trend in SSD.

The modest overall model fit ($R^2 = 0.056$) indicates that habitat type alone explains a small proportion of variation in SSD across species, suggesting that additional ecological and evolutionary factors contribute to patterns of dimorphism in Odonata.

Figure 1. Time-calibrated phylogeny of 54 dragonfly species with associated body size and sexual size dimorphism data. Branch colours indicate habitat type (orange = lentic, blue = lotic). Bar plots show male (left) and female (right) body length in centimeters. Circles indicate SSD calculated as $\log(\text{female/male body length})$, with red indicating male-biased SSD and blue indicating female-biased SSD.



Discussion

The results partially support the hypothesis that lotic dragonfly species show greater female-biased SSD than lentic species. Habitat type trended toward significance ($p = 0.084$), with lotic species consistently showing greater female-biased dimorphism. This pattern is consistent with flowing water imposing stronger ecological pressures on body size, particularly through competition for oviposition sites or fecundity selection on females.

Raw body size data showed that both sexes are substantially larger in lotic environments, but females increased slightly more than males between habitats. The greater SSD in lotic species is therefore driven primarily by proportionally larger females rather than smaller males, consistent with selection favoring larger females when reproductive resources are limited or contested.

The marginal p-value (0.084) and low R^2 (0.056) indicate that habitat type alone is a poor predictor of SSD in this dataset. The lentic/lotic dichotomy likely fails to capture the full variation in ecological pressures experienced across populations. Future work incorporating genomic estimates of effective population size (N_e) would provide a more direct test: if lotic populations experience stronger sexual selection, they should show reduced N_e relative to lentic populations, consistent with stronger directional selection. Growing dragonfly genome databases should make this feasible.

Additional limitations: 54 species is small for macroevolutionary analysis, and the Letsch et al. (2016) classifications cover only Anisoptera, excluding damselflies and limiting generalization to Odonata as a whole. Future work should expand species sampling, include damselflies, and incorporate additional variables such as habitat permanence and temperature.



Conclusions & Acknowledgements

- Habitat type trended toward predicting SSD in dragonflies after accounting for shared evolutionary history ($\lambda = 0.583$, $p = 0.084$)
- Lotic species exhibited greater female-biased SSD than lentic species, driven primarily by females growing proportionally larger than males in flowing water environments
- While the result did not reach conventional significance, the direction of the effect across analyses suggests a real biological pattern
- Future work incorporating genomic estimates of effective population size could provide a stronger test of whether lentic and lotic populations differ in the selective pressures hypothesized to drive this pattern

I would like to thank Dr. Heath Blackmon for guidance and supervision throughout this project. Body size data were sourced from Córdoba-Aguilar (2008) and habitat and phylogenetic data from Letsch et al. (2016). Statistical analyses and figure production were carried out in R using the `ape`, `caper`, `ggtree`, and `patchwork` packages. Claude (Anthropic) provided assistance with R scripting, data processing, and figure production.